

schemes through a classical linear stability analysis. Therefore, we conduct an empirical simulation-based stability study, as proposed by [27], in which simulations of the governing equations (here, the SWE on the rotating sphere), considering a steady-state solution, are performed in order to estimate the largest eigenvalue of the Jacobian of the approximate time evolution operator, which allows us to estimate the largest growth rate of perturbations to the steady state. This procedure, based on the power method for the estimation of the largest eigenvalue of a matrix, is briefly recalled below. We refer the reader to Appendix B of [27] for details and parameters to be used. Let

$$\mathbf{U}^{n+1} = \mathbf{G}(\mathbf{U}^n) \quad (7.2)$$

represent the evolution of a given state vector $\mathbf{U}$ between times t_n and t_{n+1} using a given time discretization method. Let $\bar{\mathbf{U}}$ be a steady state, such that $\mathbf{R}^n := \mathbf{U}^n - \bar{\mathbf{U}}$ is the perturbation at time n . Through linearization of (7.2), the perturbation satisfies, to first order

$$\mathbf{R}^{n+1} = \mathbf{G}'(\bar{\mathbf{U}})\mathbf{R}^n \quad (7.3)$$

As in the power method, the procedure is to iterate over (7.3). If this method converges, then $\mathbf{R}^n$ converges to the eigenvector associated with the largest (in absolute value) eigenvalue $\bar{\lambda}$ of $\mathbf{G}'$. This converged dominant eigenvalue can also be computed, as well as the growth rate $\bar{\nu} := \log \bar{\lambda} / \Delta t$ and the e-folding time $\bar{\tau}$, defined as the time in which $\mathbf{R}$ increases by a factor equal to e . We have

$$e = \frac{\|\mathbf{R}^n\|}{\|\mathbf{R}^0\|} = |\bar{\lambda}|^n = e^{\bar{\nu}n\Delta t} \implies \bar{\tau} = n\Delta t = \frac{1}{\bar{\nu}} = \frac{\Delta t}{\log \bar{\lambda}} \quad (7.4)$$

The e-folding time is a relevant parameter, since, even if the scheme is unstable (*i.e.*, $|\bar{\lambda}| > 1$), a large e-folding time may indicate that instabilities will remain reasonably small within simulation times used in practical applications, *e.g.*, in numerical weather prediction, such that the time integration scheme can be successfully used in such contexts.

We conduct this stability study by considering the geostrophic balance test case (Williamson's test case 2), to which we introduce small initial perturbations to trigger instabilities. The spectral resolution is set to $M = 512$, and we compute $\bar{\lambda}$ and $\bar{\tau}$ for various time step sizes and for the various time integration schemes considered here, with CFL numbers $c\Delta t/\Delta x$ ranging from 0.12 to 62, approximately. Results are presented in Figure 7, which only includes simulations for which the iterative procedure described above converges. The smallest reported eigenvalues are approximately 1.001. We do not observe, in this test case, clear differences in the stability behavior depending on the discretization of the linear term in the semi-Lagrangian exponential schemes, with very similar results between SE11 and SE21, and also between SE12 and SE22, *i.e.*, the stability seems to be mostly determined by the discretization of the nonlinear term. The SE12 and SE22 schemes clearly outperform the other methods, including SL-SI-SETTLS, for which the dominant eigenvalue rapidly increases when the time step sizes get larger. Also, unlikely the other schemes, SE12 and SE22 present e-folding times of more than one day for some time step sizes as large as approximately $\Delta t = 1000s$ (approximate CFL number of 8.1), indicating that simulations can be conducted for relatively large durations with instabilities remaining quite controlled. We recall that no artificial viscosity is considered in this study. Finally, we highlight the unnatural peak on the dominant eigenvalue (corresponding to the abrupt reduction of the e-folding time) of SE12 and SE22 for the tested time step sizes $\Delta t = 240s$ and $\Delta t = 480s$. As described in Section 7.4, this behavior is also observed in the simulation of the unstable jet test case, but the reasons are unknown; it may be caused, *e.g.*, by possible bad behaviors of the exponential functions when computed at time step

sizes close to the mentioned values. As identified by [9] through linear stability analyses, exponential integration methods usually have quite erratic stability behaviors, including asymmetric stability regions and discontinuities of the amplification factor as a function of the time step size.

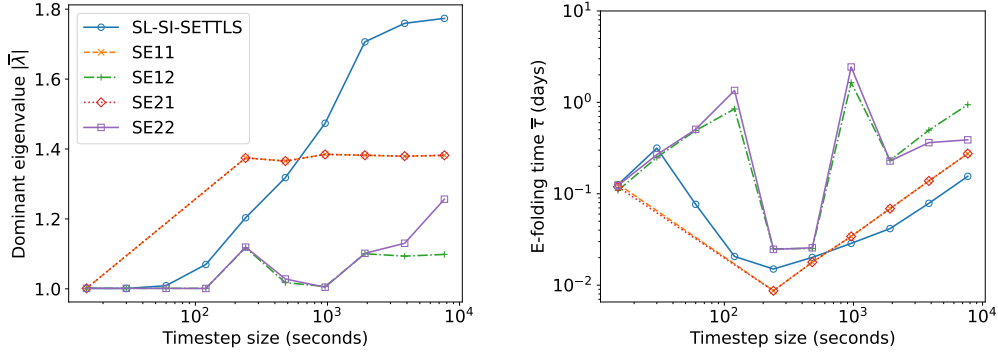


FIGURE 7. Simulation-based stability analysis considering the geostrophic balanced test case: dominant eigenvalues (left) and corresponding e-folding times (right) as a function of the time step size and the time integration scheme. The same legend holds for both plots. The smallest reported eigenvalues are approximately 1.001.

7.4. Qualitative comparison

We consider the unstable jet test case to conduct a detailed qualitative comparison of the solutions provided by SE11, SE12, their proposed modifications and SL-SI-SETTLS. This test case is standard in atmospheric modeling research due to its complex dynamics after some days of simulation, and guidelines for evaluating numerical methods applied to it have been proposed by [34]. For this study, we consider a fixed spectral resolution $M = 512$ and various time step sizes in the same range $[60, 960]$ (s) considered in the convergence analysis, with respective CFL numbers between 0.91 and 14, approximately. A reference solution is obtained from a simulation using a fourth-order explicit Runge-Kutta method, with $M = 1024$ and $\Delta t = 2$ s, corresponding to an approximate CFL number of 0.06. Results using the Eulerian exponential schemes are not considered due to their highly unstable behavior for all tested time step sizes.

We first present, in Figures 8a and 8b, the kinetic energy spectra produced after six days of simulation by the four semi-Lagrangian exponential schemes, as well as SL-SI-SETTLS, with $\Delta t = 120$ s and $\Delta t = 960$ s, and compared to the reference solution. All spectra, including the reference one, present upturned tails, indicating the accumulation of energy on the highest wavenumbers due to the enstrophy cascade, which may lead to instabilities in longer executions if there is not enough dissipation in the numerical scheme [36]; thus, an appropriate application of artificial viscosity or hyperviscosity could be considered in order to mitigate unstable behaviors.

In the case $\Delta t = 120$ s, the SE11 scheme presents a clear unstable behavior, with almost the entire spectrum being strongly overamplified. This behavior is partially controlled with the proposed modified discretization of the linear term, and SE21 presents a smaller overamplification of intermediate wavenumbers; concerning SE12 and SE22, their spectra, which almost coincide

visually, accurately reproduce the reference one in small and intermediate wavenumbers, but are damped out in the largest ones, indicating a stronger diffusion due to second-order discretization of the nonlinear term; finally, we observe an unstable behavior of SL-SI-SETTLS at the end of the wavenumber spectrum, being outperformed by the semi-Lagrangian exponential schemes using a second-order discretization of the nonlinear term.

When the larger time step size $\Delta t = 960\text{s}$ is considered, all time integration schemes present overamplifications of at least a portion of the wavenumber spectrum, notably SL-SI-SETTLS. The most stable simulation is provided by our proposed second-order scheme SE22, whose spectrum presents only a small peak at medium wavenumbers, being able to well reproduce the reference one at the large scales and remaining below it at the fine ones. Note that SE12, which provides nearly identical results to SE22 when $\Delta t = 120\text{s}$, has now a much more pronounced unstable behavior, with overamplification of almost the entire spectrum.

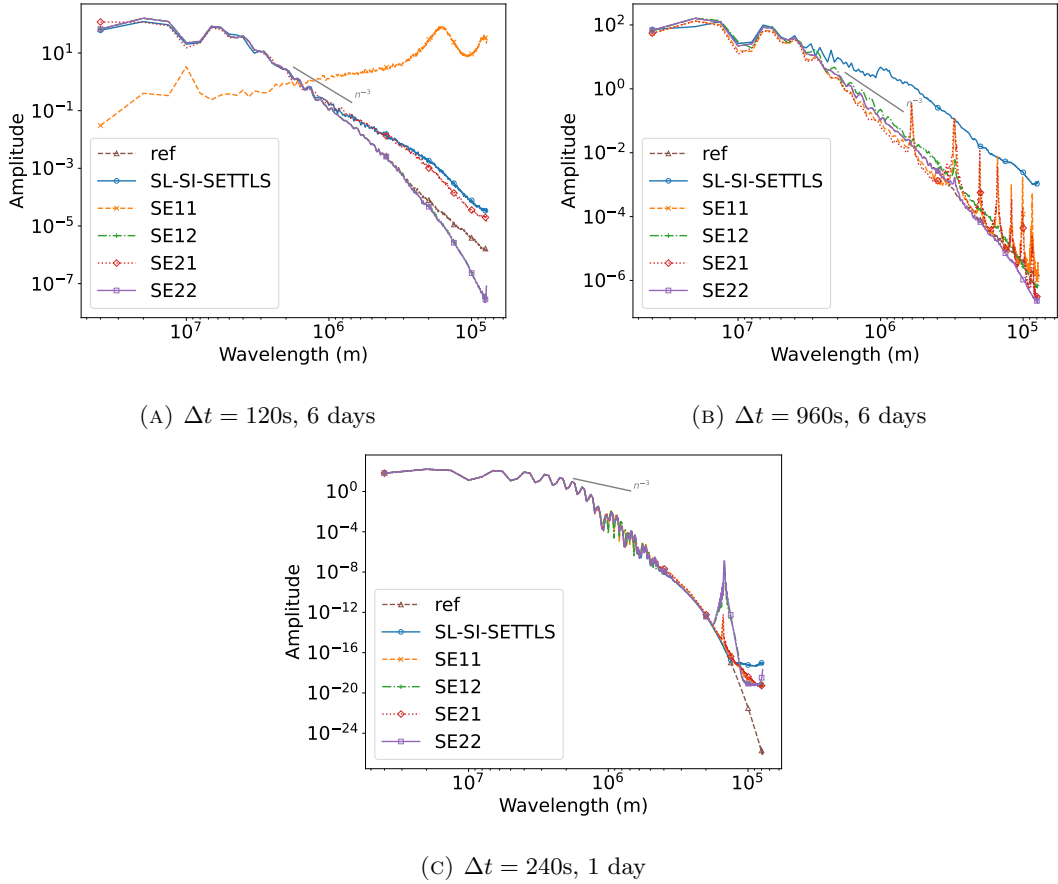


FIGURE 8. Kinetic energy spectra, truncated at resolution $M = 512$, for the unstable jet test case.